

Unit 10: Alligation or Mixture

CONTENTS

Objective

Introduction

10.1 Alligation Rule

Summary

Keywords

Self Assessment

Answers for Self Assessment

Review Questions

Further Readings

Objective

The key concepts explained in this unit, including area a learner is expected to learn and master after going through the unit are: -

- We will understand what is an Alligation.
- We will understand what are Rule of Alligation.

Introduction

Alligation literally means 'linking'. It is a rule to find:

- (a) The ratio in which two or more ingredients at their respective prices should be mixed to give a mixture at a given price.
- (b) The mean or average price of a mixture when the prices of two or more ingredients which may be mixed together and the proportion in which they are mixed are given. Here, cost price of a unit quantity of a mixture is called the 'mean price'.

10.1 Alligation Rule

1. Alligation

Alligation is the rule which enables us to find the ratio in which two or more ingredients at the given price must be mixed to produce a mixture of a specified price.

2. Mean Price

Mean price is the cost price of a unit quantity of the mixture

3. Rule of Alligation

If two ingredients are mixed, then

$$\left(\frac{\text{Quantity of cheaper}}{\text{Quantity of dearer}} \right) = \left(\frac{\text{C.P. of dearer} - \text{Mean Price}}{\text{Mean price} - \text{C.P. of cheaper}} \right)$$

The above formula can be represented with the help of the following diagram which is easier to understand.

$$\Rightarrow (\text{Cheaper quantity}) : (\text{Dearer quantity}) = (d - m) : (m - c)$$

4. Suppose a container contains x of liquid from which y units are taken out and replaced by water.

After n operations, the quantity of pure liquid $= \left[x \left(1 - \frac{y}{x} \right)^n \right]$ units.

Suppose ₹ d per unit be the price of first ingredient (superior quality) mixed with another ingredient (cheaper quality) of price ₹ c per unit to form a mixture whose mean price is ₹ m per unit, then the two ingredients must be mixed in the ratio:

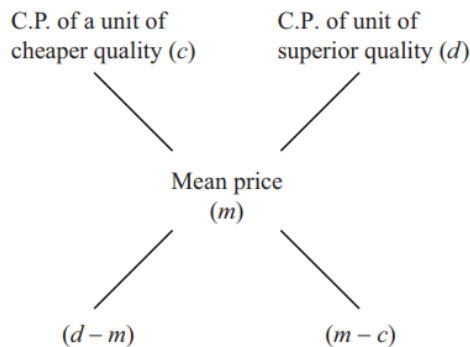
i.e., the two ingredients are to be mixed in the inverse ratio of the differences of their prices and the mean price. The above rule may be represented schematically as under:

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$$\frac{\text{Quantity of cheaper}}{\text{Quantity of superior}} = \frac{\text{C.p. superior} - \text{Mean price}}{\text{Mean price} - \text{C.P. of cheaper}}$$

i.e., the two ingredients are to be mixed in the inverse ratio of the differences of their prices and the mean price. The above rule may be represented schematically as under:



$$\frac{\text{Quantity of cheaper quality}}{\text{Quantity of superior quality}} = \frac{d-m}{m-c}$$

Explanation:- Suppose, x Kg of cheaper quality is mixed with y Kg of superior quality.

Price of cheaper ingredient = ₹ cx

Price of superior ingredient = ₹ dy

$$\therefore \text{Price of mixture} = ₹ \left(\frac{cx+dy}{x+y} \right)$$

and quantity of mixture = $(x + y)$ Kg.

$$\therefore \frac{cx+dy}{x+y} = m$$

$$cx + dy = mx + my$$

$$\Rightarrow dy - my = mx - cx$$

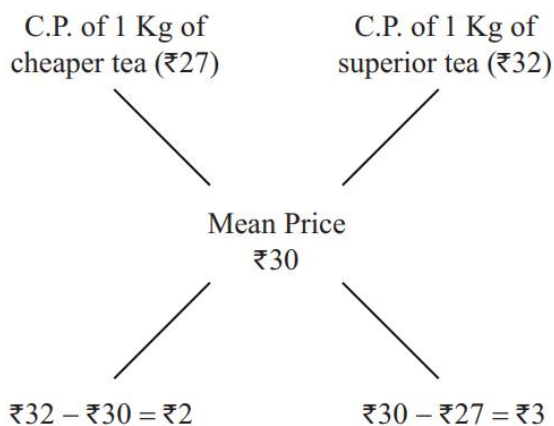
$$\Rightarrow y(d - m) = x(m - c)$$

$$\Rightarrow \frac{x}{y} = \frac{d-m}{m-c}$$



Example:- 1: In what ratio two varieties of tea, one costing ₹ 27 per Kg and the other costing ₹ 32 per Kg, should be blended to produce a blended variety of tea worth ₹ 30 per Kg. How much should be the quantity of second variety of tea, if the first variety is 60 Kg.

Solution:



The required ratio of the two varieties of tea is 2:3, i.e.,

$$\frac{\text{Quantity of cheaper tea}}{\text{Quantity of superior tea}} = \frac{2}{3}$$

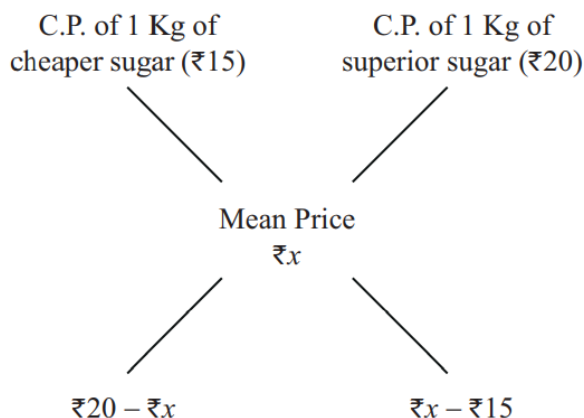
$$\therefore \text{Quantity of superior tea} = \frac{60 \times 3}{2} = 90 \text{ kg}$$

Thus, the second variety of tea is 90 Kg.



Example:- 2: Sugar at ₹ 15 per Kg is mixed with sugar at ₹ 20 per Kg in the ratio 2:3. Find the per Kg price of the mixture.

Solution: Let, the mean price of the mixture be ₹ x



$$\frac{\text{Quantity of cheaper sugar}}{\text{Quantity of dearer sugar}} = \frac{20 - x}{x - 15}$$

$$\Rightarrow 60 - 3x = 2x - 30$$

$$\Rightarrow 5x = 90 \text{ or, } x = 18.$$

Thus, the per Kg price of the mixture is ₹18.

01) A vessel, full of wine, contains a litres of it of which b litres are withdrawn. The vessel is then filled with water. Next, b litres of the mixture are withdrawn and, again the vessel is filled with water. This process is repeated n times. Then

$$\frac{\text{wine left in the vessel after } n\text{th operations}}{\text{original quantity of wine in the vessel}}$$

$$= \left(\frac{a-b}{a} \right)^n$$

Explanation:-

Amount of wine after the first operation

$$= a-b = \left(1 - \frac{b}{a}\right) \times a$$

Ratio of wine and water after the first operation is $(a - b) : b$.

$\therefore$ In b litres of mixture withdrawn in the second operation, amount of wine withdrawn

$$= \frac{a-b}{(a-b)+b} \times b = (a-b) \frac{b}{a}$$

$\therefore$ Amount of wine left after the second operation

$$= (a-b) - (a-b) \times \frac{b}{a} = \left(1 - \frac{b}{a}\right)^2 a$$

In general, quantity of wine left after n th operation $= \left(1 - \frac{b}{a}\right)^n a$

$$\therefore \frac{\text{wine left after nth operation}}{\text{original quantity of wine}} = \left(1 - \frac{b}{a}\right)^n$$



Example 3: Two salt solutions have concentrations 15% and 40%. In what ratio should they be mixed to get a solution of concentration 30%?

Solution:

So, they should be mixed in the ratio 2: 3.



Example 4:- X and Y two alloys are made by mixing aluminum and magnesium metals in the ratio of 8: 5 and 9:16 respectively. If equal amounts of alloys are melted to form a new alloy Z, what will be the ratio of aluminum and magnesium in Z?

- a) 317:333
- b) 316:319
- c) 314:333
- d) 313:317

Answer: a

Solution:

The ratio of aluminum and magnesium in Z = $[8/13+9/25] : [5/13+16/25]$

$$= 317/325 : 333/325$$

$$= 317:333$$



Example 5:- A cistern contains 100 liters of water. 10 liters of water is taken out of it and replaced by the same quantity of soda. This process is repeated one more time. After that 20 liters of the solution is replaced by the same quantity of whiskey. Find the proportion of soda, water, and whiskey in the final mixture?

- a) 19: 71: 25
- b) 38: 72: 50
- c) 19: 81: 25
- d) 29: 71: 25

Answer: c

Solution:

Given

A cistern contains 100 liters of water.

Calculation

Initially

Water = 100 lit

After 1st replacement

Water: Soda = 90: 10 = 9: 1

After 2nd replacement

Water = $90 - (10 \times 9/10) = 81$ lit

Soda = $10 - (10 \times 1/10) + 10 = 19$ lit

After 3rd replacement

Water = $81 - (20 \times 81/100) = 64.8$ lit

Soda = $19 - (20 \times 19/100) = 15.2$ lit

Whiskey = 20 lit

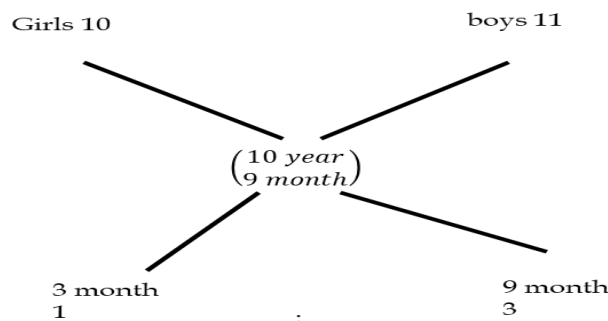
∴ Required ratio = 15.2: 64.8: 20

⇒ 19: 81: 25



Example 6:- There are 400 students in a Mathematics class. Average age of girls is 10 years and average age of boys is 11 years. Find the number of girls in the class, if the average age of all the students is 10 years 9 months.

Solutions:-



So, number of girls = $1/4 \times 400 = 100$



Example 7:- A vessel contains 125 litres of wine. 25 litres of wine was taken out of the vessel and replaced by water. Then, 25 litres of mixture was withdrawn and again replaced by water. The operation was repeated for the third time. How much wine is now left in the vessel? Solution: Amount of wine left in the vessel

$$= \left(1 - \frac{25}{125}\right)^3 \times 125$$

$$= \frac{100 \times 100 \times 100 \times 125}{125 \times 125 \times 125}$$

= 64 Liters

There are n vessels of equal size filled with mixtures of liquids A and B in the ratio $a_1:b_1, a_2:b_2, \dots, a_n:b_n$, respectively. If the contents of all the

vessels are poured into a single large vessel, then

$$\frac{\text{Quantity of liquid A}}{\text{Quantity of liquid B}} = \frac{\left(\frac{a_1}{a_1+b_1} + \frac{a_2}{a_2+b_2} + \dots + \frac{a_n}{a_n+b_n}\right)}{\left(\frac{b_1}{a_1+b_1} + \frac{b_2}{a_2+b_2} + \dots + \frac{b_n}{a_n+b_n}\right)}$$

Explanation:- Let the capacity of each vessel be c litres. Amount of liquid A in different vessels

$$= \frac{a_1 c}{a_1 + b_1}, \frac{a_2 c}{a_2 + b_2}, \frac{a_3 c}{a_3 + b_3}, \dots, \frac{a_n c}{a_n + b_n}$$

Amount of liquid B in different vessels

$$= \frac{b_1 c}{a_1 + b_1}, \frac{b_2 c}{a_2 + b_2}, \frac{b_3 c}{a_3 + b_3}, \dots, \frac{b_n c}{a_n + b_n}$$

So, in the resulting mixture, amount of liquid A

$$= \left(\frac{a_1}{a_1 + b_1} + \frac{a_2}{a_2 + b_2} + \dots + \frac{a_n}{a_n + b_n} \right) \times c$$

Amount of liquid B

$$= \left(\frac{b_1}{a_1 + b_1} + \frac{b_2}{a_2 + b_2} + \dots + \frac{b_n}{a_n + b_n} \right) \times c$$

$$\frac{\text{Quantity of liquid A}}{\text{Quantity of liquid B}} = \frac{\left(\frac{a_1}{a_1 + b_1} + \frac{a_2}{a_2 + b_2} + \dots + \frac{a_n}{a_n + b_n} \right)}{\left(\frac{b_1}{a_1 + b_1} + \frac{b_2}{a_2 + b_2} + \dots + \frac{b_n}{a_n + b_n} \right)}$$



Example 8:- How many kilograms of sugar of Rs.5.4 per kg should be mixed with 10 kg of sugar of Rs.4.5 per kg, such that there may be gain of 20% by selling the mixture at Rs.5.94 per kg.

- a) 10 kg
- b) 12 kg
- c) 15 kg
- d) 8 kg

Answer: A

Solution:

Let, the amount of rice of Rs.5.4 per kg = x kg

According to the question,

$$x \times 5.4 + 4.5 \times 10 = 5.94 \times (10 + x) \div 120 \times 100$$

$$= 5.4x + 45 = 4.95 \times (10 + x)$$

$$= 5.4x + 45 = 49.5 + 4.95x$$

$$= 5.4x - 4.95x$$

$$= 49.5 - 45$$

$$= 0.45x = 4.5$$

$$= x = 10 \text{ kg}$$



Example 9:- Three equal glasses are filled with mixture of milk and water. The proportion of milk and water in each glass is as follows: In the first glass as 3:1, in the second glass as 5:3 and in the third as 9:7. The contents of the three glasses are emptied into a single vessel. What is the proportion of milk and water in it?

Solution:

$$\frac{\text{Quantity of milk}}{\text{Quantity of water}} = \frac{\frac{3}{3+1} + \frac{5}{5+3} + \frac{9}{9+7}}{\frac{1}{3+1} + \frac{3}{5+3} + \frac{7}{9+7}}$$

$$= \frac{31/16}{17/16}$$

$$= 31:17$$

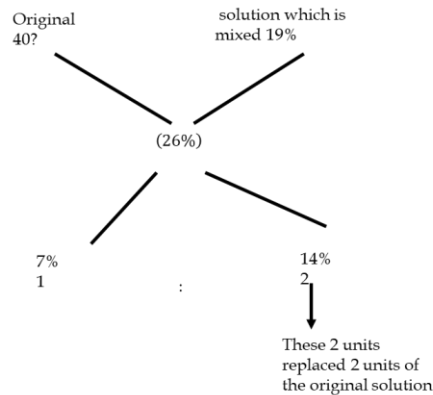
There are n vessels of sizes $c_1, c_2, \dots, c_n$ filled with mixtures of liquids A and B in the ratio $a_1 : b_1, a_2 : b_2, \dots, a_n : b_n$, respectively. If the contents of all the vessels are poured into a single large vessel, then

$$\frac{\text{Quantity of milk}}{\text{Quantity of water}} = \frac{\frac{a_1}{a_1+b_1} + \frac{a_2}{a_2+b_2} + \dots + \frac{a_n}{a_n+b_n}}{\frac{b_1}{a_1+b_1} + \frac{b_2}{a_2+b_2} + \dots + \frac{b_n}{a_n+b_n}}$$



Example 10:- Some part of a sugar solution which contains 40% sugar is replaced with another sugar solution which contains 19% sugar. Part of sugar in the new mixture became 26% what fraction of the original sugar solution was replaced with another sugar solution?

Solutions:



Total number of units = 1 + 2 = 3

2 units of original solution, out of 3, were replaced.

= 2/3 of the original solution was replaced.



Example 11:- Three glasses of sizes 3 litres, 4 litres and 5 litres, contain mixture of milk and water in the ratio 2:3, 3:7 and 4:11, respectively. The contents of all the three glasses are poured into a single vessel. Find out ratio of milk to water in the resulting mixture

Solution : $\frac{\text{quantity of milk}}{\text{quantity of water}}$

$$\begin{aligned} &= \frac{\left(\frac{2 \times 3}{2+3} + \frac{3 \times 4}{3+7} + \frac{4 \times 5}{4+11}\right)}{\left(\frac{3 \times 3}{2+3} + \frac{7 \times 4}{3+7} + \frac{11 \times 5}{4+11}\right)} \\ &= \frac{\frac{6}{5} + \frac{12}{10} + \frac{20}{15}}{\frac{9}{5} + \frac{28}{10} + \frac{55}{15}} \end{aligned}$$

= 56:124

OR 14: 31



Example 12:- A jar had 330-liter mixture of Milk and water in the respective ratio of 2 : 1.60 liters of this mixture is taken out and 'x' liter each of milk and water is added to the jar (remaining mixture). The respective ratio between milk and water was 8 : 5 respectively. What was the total quantity of both milk and water added to the jar?

- a) 90 liter
- b) 170 liter
- c) 160 liter
- d) 120 liter

Answer: d

Solution:

Analytical Skills-I

Given

Total mixture = 330

liter Ratio of milk and water = 2 : 1

Calculation

Mixture left in jar after 60 liters taken out = $330 - 60 = 270$ liters

Milk = $(2/3) \times 270 = 180$ liter

Water = $(1/3) \times 270 = 90$ liter

Now according to question

$$(180 + x) / (90 + x) = 8/5$$

$$\Rightarrow 900 + 5x = 720 + 8x$$

$$\Rightarrow 3x = 180$$

$$\Rightarrow x = 60$$

$\therefore$ Quantity that was added = $2 \times 60 = 120$ liter.



Example 13:- A container contains two liquids A and B in the ratio of 3 : 2. When 8 L of liquid B is added to X L of the mixture, the ratio of liquid A to liquid B becomes 1 : 1. What is the value of X?

a) 50 L

b) 45 L

c) 40 L

d) None of these

Ans: c)

Sol:

Given, Ratio of A and B in the mixture 3 : 2

Volume of B added = 8 L

Final ratio of A and B in the mixture = 1 : 1

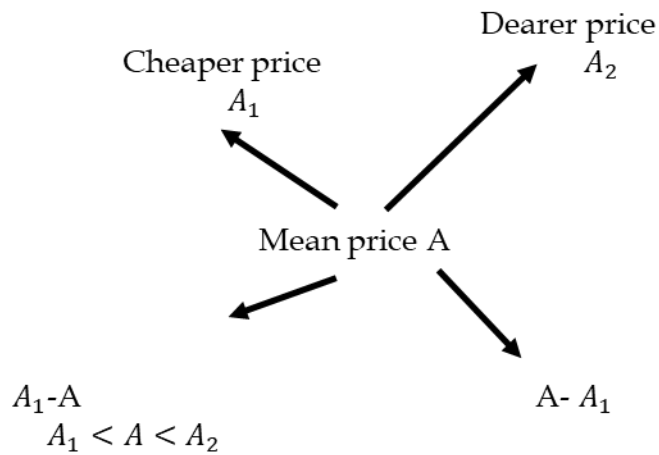
Formula used,

To find the ratio in which two Quantities are mixed to form a new mixture is as shown below.

Where A_1 , A_2 and A are the cheaper price, Dearer price and Mean price, respectively. Then ratio of cheaper Quantity to Dearer quantity is calculated as ($n_1 : n_2$)

$$n_1 : n_2 = (A_2 - A) / (A - A_1)$$

Also, the Mean value should lie between Cheaper price and Dearer price



applying Alligation concept

Proportion B in the mixture = $2/5$ (cheaper)

Pure liquid B = 1 (dearer)

Proportion of B in final mixture = $1/2$ (mean)

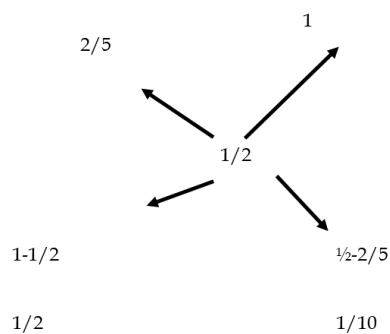
Using Alligation concept Ratio of cheaper to dearer = $1/2 : 1/10 = 5 : 1$

1R refers to pure liquid of B = 8 L

$\therefore 1R = 8 \text{ L}$

$\therefore 5R = 40 \text{ L}$

Therefore, X = 40 L



Example 14:- In what proportion a milkman must mixed water with milk, so that after selling the mixture at half the cost of milk, gains 40%

a) 9 : 5

b) 5 : 9

c) 4 : 5

d) 3 : 5

Ans: a

Sol:

Given,

Cost of mixture = $1/2 \times$ cost of milk

Percentage profit = 40%

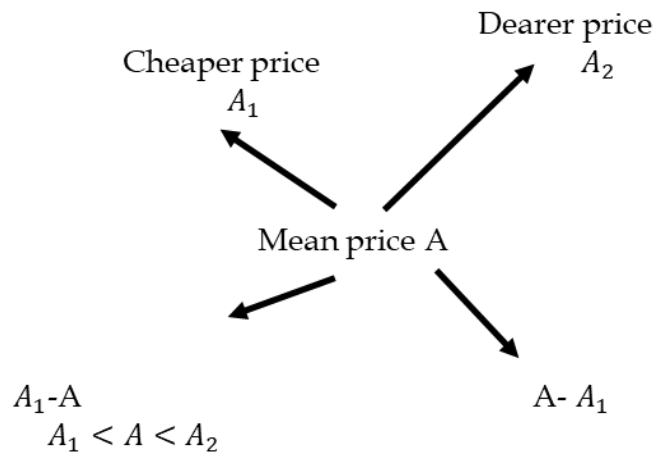
Formula used,

To find the ratio in which two Quantities are mixed to form a new mixture is as shown below.

Where A_1 , A_2 and A are the cheaper price, Dearer price and Mean price, respectively. Then ratio of cheaper Quantity to Dearer quantity is calculated as

($n_1 : n_2$)

$n_1 : n_2 = (A_2 - A) / (A - A_1)$ Also the Mean value should lie between Cheaper price and Dearer price



Let cost of milk = Rs. X per litre

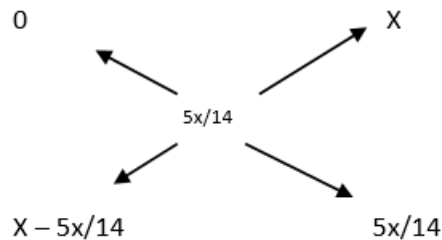
$\therefore$ Selling price of mixture of milk and water = Rs. $X/2$ per litre (dearer)

Cost of water = Rs. 0 per litre (cheaper)

Percentage profit after selling the mixture = 40%

$\therefore$ Cost price of the mixture = $X/2 \times 100/140 = 5X/14$ (mean)

Applying Alligation rule:-



Ratio of cheaper to dearer = $9X/14 : 5X/14 = 9 : 5$

Therefore, ratio of water and milk = $9 : 5$



Example 15:- Ratio of Milk and water in Vessel A is 7:4 and same mixture in the ratio of 5:3, 22 liters mixture from vessel A taken out and poured in vessel B new ratio of milk to water is 27:16. If new quantity of mixture in vessel B is equal to initial quantity of mixture in vessel A, then find quantity of Milk after 8 liters of mixture has been taken out from Vessel A?

- A) 200/29 liters
 B) 546 / 11 liters
 C) 120/13 liters
 D) 220/19 liters

Answer: b

Let Ratio of Milk and water in Vessel A is 7x and 4y

Let ratio of milk and water in vessel B is 5y and 3y

Now according to question.

$$(5y + 22 * 7 / 11) / (3y + 22 * 4 / 11) = 27 / 16$$

$$(5y + 14) / (3y + 8) = 27 / 16$$

$$80y + 224 = 81y + 216$$

$$Y = 8$$

New quantity of mixture in vessel B

$$=> (8 * 5 + 14) + (8 * 3 + 8)$$

$$=> 54 + 32$$

$$=> 86$$

Therefore, initial quantity of vessel A = 86 liters

Quantity of milk remaining in Vessel A

$$86 * 7 / 11 - 8 * 7 / 11$$

$$=> 546 / 11 \text{ litres.}$$



Example 16:- A mixture contains 40% milk and another mixture contains milk and water in the ratio of 1: 5. How many litres of the latter must be mixed with 6 litres of the former so that the resulting mixture may contain milk and water in the ratio of 1: 2.

- a) 10 L
 b) 15 L
 c) 18 L
 d) 20 L

Ans: b

Sol: Given,

Ratio of milk and water in 1st mixture = 40 : 60 or 2 : 3

Ratio of milk and water in 2nd mixture = 1 : 5

Quantity of 1st mixture = 6 L

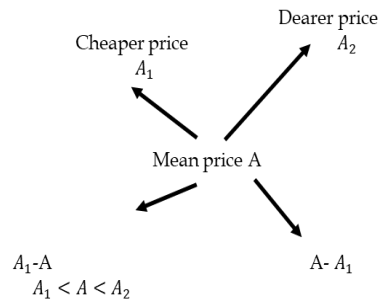
Ratio of milk and water in final mixture = 1 : 2

Formula used,

To find the ratio in which two Quantities are mixed to form a new mixture is as shown below.

Where A1, A2 and A are the cheaper price, Dearer price and Mean price, respectively.

Then ratio of cheaper Quantity to Dearer quantity is calculated as $(n1 : n2) \quad n1 : n2 = (A2 - A) / (A - A1)$ Also the Mean value should lie between Cheaper price and Dearer price

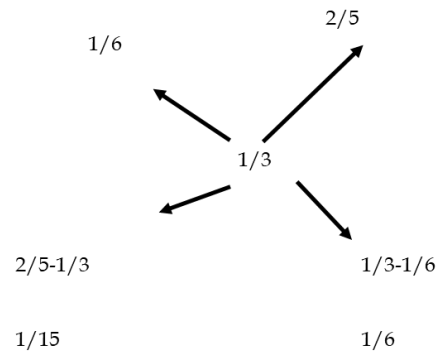


Proportion of milk in 1st mixture = $2/5$ (Dearer value)

Proportion of milk in 2nd mixture = $1/6$ (Cheaper value)

Proportion of milk in final mixture = $1/3$ (Mean value)

Applying Alligation rule,



Cheaper : Dearer = $1/15 : 1/6 = 2 : 5$

2R refers to former quantity of mixture

3R refers to latter quantity of mixture

$\therefore 2R = 6 \text{ L} \therefore 5R = 15 \text{ L}$

Therefore, Quantity of latter mixture = 15 L .



Example 17:- Three containers of equal capacity are containing a mixture of wine and water in the ratio of 2 : 1, 5 : 4 and 3 : 4 respectively. These three containers are emptied into a 4th container. What is the approximate percentage of wine in 4th container?

a) 50%

b) 60%

c) 45%

d) 55%

Ans: d

Sol:

Given,

Ratio of wine and water in 1st container = 2 : 1 Ratio of wine and water in 2nd container = 5 : 4 Ratio of wine and water in 3rd container = 3 : 4

$1/6 \quad 2/5$

$1/3$

$2/5 - 1/3 \quad 1/3 - 1/6$

$1/15 \quad 1/6$

Proportion of wine in 1st container = $\frac{2}{3}$ and proportion of water = $\frac{1}{3}$

Proportion of wine in 2nd container = $\frac{5}{9}$ and proportion of water = $\frac{4}{9}$

Proportion of wine in 3rd container = $\frac{3}{7}$ and proportion of water = $\frac{4}{7}$

Required ratio of wine and water in 4th container is

$$(\frac{2}{3} + \frac{5}{9} + \frac{3}{7}) : (\frac{1}{3} + \frac{4}{9} + \frac{4}{7})$$

$$104 : 85$$

$$\text{Percentage of wine} = \frac{104}{189} \times 100 = 55\%$$

Summary

The key concepts learned from this unit are: -

- We have learnt what is an Alligation.
- We will understand what are Rule of Alligation.

Keywords

- Alligation.
- Mean price.
- Mean
- Average
- Ratio

Self Assessment

1. Two varieties of sugar are mixed in the proportion of 4:7 and the mixture is sold at Rs. 39 per kg at a profit of 30%. If the second variety of sugar cost Rs. 5.5 more than that of first variety of sugar. Find the cost price of the first variety of sugar.
A. 31 Rs/kg
B. 35.5 Rs/kg
C. 37.5 Rs/kg
D. None of these
2. How many kg of salt at 42 paise per kg must a man mix with 25 kg of salt at 24 paise per kg, so that he may, on selling the mixture at 40 paise per kg. gain 25% on the outlay?
A. 20
B. 30
C. 40
D. 50
3. The ratio of the quantities of acid and water in a mixture is 1:3. If 5 litres of acid is further added to the mixture, the new ratio becomes 1:2. The quantity of new mixture in litres is
A. 32
B. 40
C. 42
D. 45
4. A barrel contains a mixture of wine and water in the ratio 3: 1. How much fraction of the mixture must be drawn off and substituted by water so that the ratio of wine and water in the resultant mixture in the barrel becomes 1: 1?
A. $\frac{1}{4}$
B. $\frac{1}{3}$

Analytical Skills-I

- C. $\frac{2}{3}$
D. $\frac{3}{2}$
5. A mixture contains spirit and water in the ratio 3 : 2. If it contains 3 litres more spirit than water, the quantity of spirit in the mixture is
A. 10 litres
B. 12 litres
C. 8 litres
D. 9 litres
6. There are 81 litres pure milk in a container. One-third of milk is replaced by water in the container. Again one-third of mixture is extracted and equal amount of water is added. What is the ratio of milk to water in the new mixture?
A. 1: 2
B. 1: 1
C. 2: 1
D. 4: 5
7. Three vessels whose capacities are 3:2:1 are completely filled with milk mixed with water. The ratio of milk and water in the mixture of vessels are 5:2, 4:1 and 4:1 respectively. Taking $\frac{1}{3}$ of first, $\frac{1}{2}$ of second and $\frac{1}{7}$ of third mixtures, a new mixture, kept in a new vessel is prepared. The percentage of water in the new mixture is
A. 32
B. 28
C. 30
D. 24
8. Three containers whose volumes are in the ratio of 2:3:4 are full of mixture of spirit and water. In the 1st container the ratio of spirit and water is 4:1 in 2nd container the ratio is 11:4 and in the 3rd container ratio is 7:3. All the three mixtures are mixed in a big container. The ratio of spirit and water in the resultant mixture is:
A. 4:9
B. 9:5
C. 11:4
D. 5:10
9. Bottle 1 contains a mixture of milk and water in 7:2 ratio and Bottle 2 contains a mixture of milk and water in 9:4 ratio. In what ratio of volumes should the liquids in Bottle 1 and Bottle 2 be combined to obtain a mixture of milk and water in 3:1 ratio?
A. 27:14
B. 27:13
C. 27:16
D. 27:18
10. A milk vendor sells 10 litres of milk from a can containing 40 litres of pure milk to the 1st customer. He then adds 10 litres of water to the milk can. He again sells 10 litres of mixture to the 2nd customer and then adds 10 litres of water to the can. Again, he sells 10 litres of mixture to the 3rd customer and then adds 10 litres of water to the can and so on. What amount of pure milk will the 5th customer receive?
A. $(\frac{510}{128})$ Litres
B. $(\frac{505}{128})$ Litres
C. $(\frac{410}{128})$ Litres
D. $(\frac{405}{128})$ Litres
11. Two liquids A and B are mixed in the ratio of 1 : 4 and the mixture is sold at Rs.46 per litre at a profit of 15%. If the liquid A costs Rs. 10 less than the cost of B. cost of B is (In Rs.)
A. 40
B. 42
C. 50
D. 32

Unit 10: Alligation or Mixture

12. A can contains a mixture of two liquids A and B in the ratio of 7: 9. When X litres of mixture is removed and is replaced by B. The ratio of A and B becomes 21: 43. Find X is how much percent of the initial mixture.

- A. 25%
- B. 20%
- C. 35%
- D. 30%

13. There is 128 L of pure acid in a container. $\frac{1}{4}$ th of acid is removed and replaced by water. This process is repeated further 2 more times. Find the difference of acid and water in the final mixture.

- A. 40 L
- B. 20 L
- C. 30 L
- D. Can't be determined

14. Two containers A and B contains milk and water in the ratio of 2 : X and 4 : 1. These two containers are mixed in the ratio of 3 : 1, making mixture half milk and half water. Find X

- A. 5
- B. 7
- C. 3
- D. None of these

15. A milkman bought 20 L of milk and mixed 4 L water in it. If the cost of mixture becomes Rs. 30 per litre. What is the cost of milk per litre?

- A. Rs. 40
- B. Rs. 38
- C. Rs. 36
- D. Rs. 35

Answers for Self Assessment

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|-------|-------|-------|-------|-------|
| 1. D | 2. A | 3. D | 4. B | 5. B |
| 6. D | 7. D | 8. C | 9. B | 10. D |
| 11. B | 12. A | 13. B | 14. C | 15. C |

Review Questions

- Two containers A and B contains milk and water in the ratio of 2 : X and 4 : 1. These two containers are mixed in the ratio of 3 : 1, making mixture half milk and half water. Find X.
- There is 128 L of pure acid in a container. $\frac{1}{4}$ th of acid is removed and replaced by water. This process is repeated further 2 more times. Find the difference of acid and water in the final mixture.
- A can contains a mixture of two liquids A and B in the ratio of 7: 9. When X litres of mixture is removed and is replaced by B. The ratio of A and B becomes 21: 43. Find X is how much percent of the initial mixture.
- A Jar 'A', 140 litre milk was mixed with 40 litre water. Some of this mixture was taken out from Jar 'A' and put in Jar 'B'. If before the operation, there was 17 litres of milk in Jar 'B', and afterwards the resultant ratio between milk and water in Jar 'B' was 19:3 respectively, what was the amount of mixture that was taken out from Jar 'A'? (in litre)
- In Jar A, 180 litre milk was mix with 36 litre water. Some of this mixture was taken out from Jar A and put it in Jar B. If after adding 6 litres of water in the mixture, the respective

Analytical Skills-I

- ratio between milk and water in Jar B was 5:2 respectively, what was the amount of mixture that was taken out from Jar A? (in litres)
6. A vessel contains a mixture of grape, pineapple and banana juices in the respective ratio of 4: 6: 5. 15 litres of this mixture is taken out and 8 litres of grape juice and 2 litres of pineapple juice is added to the vessel. If the resultant quantity of grape juice is 10 litres less than the resultant quantity of pineapple juice, what was the initial quantity of mixture in the vessel?
 7. Suresh is having three brands of whisky with him White Dog, Renders Pride, MAT49 having ratio of alcohol to water 2:3, 5:7, 6:11 respectively. If he mixes equal quantity of each brand. Find the ratio of water to alcohol in the resultant mixture?
 8. Container P contains V litres of pure water. Container Q contains V litres of pure milk. X litres are taken out from container P and poured into container Q. Now, X litres are taken from container Q and poured into container P. Which of the following statement is true?
 - I. % of milk in container P = % of water in Container Q
 - II. % of water in container P = % of milk in Container Q
 - III. % of milk in container Q < % of water in container P
 - IV. % of milk in container P > % of water in container Q
 9. In what ratio must a person mix three kinds of wheat costing him 1.20, 1.44 and 1.74 per kg, so that mixture may worth 1.41 per kg?
 10. Two varieties of sugar are mixed in the proportion of 4:7 and the mixture is sold at Rs. 39 per kg at a profit of 30%. If the second variety of sugar cost Rs. 5.5 more than that of first variety of sugar. Find the cost price of the first variety of sugar.

**Further Readings**

1. Quantitative Aptitude For Competitive Examinations By Dr. R S Aggarwal, S Chand Publishing
2. A Modern Approach To Verbal & Non-Verbal Reasoning By Dr. R S Aggarwal, S Chand Publishing
3. Magical Book On Quicker Maths By M Tyra, Banking Service Chronicle
4. Analytical Reasoning By M.K. Pandey, Banking Service Chronicle